

SIGHT Project - Meeting 21

Project Name: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers

Project Number: WSIC26-250206-009

Date: June 26, 2026

Time: 11:00 AM EDT

Location: Zoom

Meeting Focus: Administrative updates, NAMRC 54 recap, BWC Q4 presentation recap, MU digital twin technical updates, and Discovery Center evaluation overview.

Attendees

- **Miami University (MU):** Fadel Megahed (PI), Jay Shan (Co-PI), Ibrahim Yousif (Post Doc), Michael Wise (GRA), Mohamed Farrag (Project Manager), Leo Lyublinski (Student Assistant)
- **Discovery Center:** Kristen Morio (Evaluator), Yue Li (Evaluator)
- **Consultant:** Lora Cavuoto
- **Maxbyte:** Naveen Karthi (Engineer)

(Note: Industry partners were not required to present updates during this short session).

1. Introductory Remarks & Recap

Welcome & Recap: Fadel Megahed opened the meeting, noting it would be a brief session primarily focused on MU and Discovery Center updates.

- **MeetKai Update:** MeetKai has successfully transferred all project assets (digital environment, assets, and code) to a Miami University GitHub repository.
- **MaxByte Update:** MaxByte held a demonstration at the AM Hub earlier in the week for industry participants, showcasing their work on the autonomous pallet truck and AR components. MU team (Ibrahim and Michael) supplemented this by sharing IoT integration and digital twin progress.

2. Administrative Updates

- **BWC Deliverables & Q4 Recap:**
 - The Q4 BWC presentation was submitted on June 10 and presented on June 18. The presentation highlighted the project's unique two-way communication capabilities within the digital twin.

- A follow-up meeting was held on Tuesday with BWC leadership (Glenn, Stacy, Daniel, Zach) to share live data and detailed videos of the digital twin.
- The BWC has officially approved all Q1-Q3 submissions.
- **Upcoming Deadline:** The Interim (Full-Year) Progress Report is due July 17, 2026. The MU team will dedicate the first 10 days of July to documenting progress. Yue Li will review the BWC template to integrate the Discovery Center's evaluation appropriately.
- **NAMRC 54 Recap:** Fadel and Ibrahim successfully presented their chatbot paper at NAMRC 54, generating significant interest from attendees, particularly regarding how the chatbot supports the broader AR/VR environment.
- **HICSS Conference Submission:** On June 14, the team submitted a 10-page paper to the HICSS (Hawaii International Conference on System Sciences) fast track. The paper evaluates LLM context window generation (inputting 300k-word manuals directly) versus RAG (Retrieval-Augmented Generation). Findings showed an 8% increase in "epistemic accuracy" but at a 26x higher token cost.

3. Technical Updates: MU

Digital Twin (DT) & Computer Vision Progress:

Ibrahim Yousif and Michael Wise presented the latest developments in the digital twin and computer vision environments:

- **Environment Rendering:** The digital twin floors have been rendered to match the colors and themes of the physical AM Hub facility and the web-based VR.
- **Physics & Tracking:** 3D car parts are being placed on workstations with physics applied to allow for real-time tracking and movement visualization.
- **CNC Integration:** Digital modeling for the CNC machine is underway, with the next step being connecting the CNC controller directly to Isaac Sim.
- **Camera Simulation:** The team is simulating the physical top-down camera view within Isaac Sim to match real-world monitoring angles.
- **Smart Safety Dashboard:** A computer vision dashboard was demonstrated that detects human presence, tracks time spent at stations, and monitors PPE compliance (e.g., hard hats). It includes functionality to export daily safety reports to PDF or Excel. Next steps include refining the system to count detected objects rather than just recording safety incident instances.
- **Version Control Integration:** Fadel requested that all computer vision and UI code be placed in a shared version control repository (GitHub). This will allow us to optimize the UI/UX collaboratively.

4. Project Evaluation: Discovery Center

- **Evaluation Status:** Kristen Morio and Yue Li provided a brief update on the qualitative evaluation. They invited 15-20 project participants for interviews and completed 7.
- **Next Steps:** The Discovery Center is currently synthesizing the data to identify challenges and facilitators. They aim to have a draft of the evaluation report ready by July 8th or 9th for inclusion in the BWC Interim Report.

5. Wrap-Up & Looking Ahead

- The next bi-monthly meeting will be held on **July 10, 2026**.
- A significant portion of the July 10 meeting will be dedicated to reviewing the draft of the BWC Interim Progress Report.

6. Action Items

Task	Responsible	Deadline
Submit proof of Q4 attendance to BWC	Fadel / MU Team	Done
Provide draft evaluation report for BWC inclusion	Kristen & Yue (Discovery Center)	July 8-9, 2026
Review the Interim Report Draft at the next meeting	All Project Members	July 10, 2026
Submit Interim (Full Year) Progress Report	Fadel / MU Team	July 17, 2026
Establish shared version control (GitHub/GitLab) for CV/UI/DT code	Ibrahim	ASAP
Update digital avatars/assets to wear long pants (industrial compliance)	Ibrahim / MU Team	Next Phase
Explore part tracking via YOLO model for cycle time/productivity metrics	Ibrahim / Michael	Next Phase